

If MgCO_3 undergoes a double replacement reaction in 6 M HCl, and one of the resulting products then undergoes a decomposition reaction, the gas that evolves is:

- ☐ A. hydrogen. [27%]
- ✔ ☒ B. carbon dioxide. [67%]
- ☐ C. oxygen. [3%]
- ☐ D. magnesium oxide. [1%]

Correct

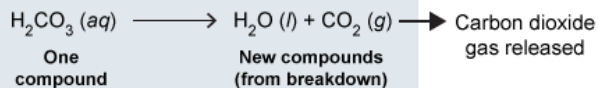
66% answered correctly

Explanation:

Double replacement



Decomposition



©UWorld

Double replacement reactions involve an **exchange** of constituent **bonding partners** between two reacting compounds and result in the formation of two new product compounds. For **double replacement reactions** between two ionic species, the cations and anions of the two compounds exchange counterions. **Decomposition reactions** follow the pattern of **one reactant** species **breaking down** (decomposing) into two or more product species. Applying **bonding patterns** for recognized **reaction types** allows the prediction of reaction products.

If MgCO_3 undergoes a double replacement reaction in 6 M HCl , the Mg^{+2} and CO_3^{-2} ions in MgCO_3 exchange bonding partners with the H^+ and Cl^- ions in HCl (cations pairing with anions). This causes the formation of water-soluble MgCl_2 (an ionic salt) and H_2CO_3 (an unstable acid). The unstable H_2CO_3 undergoes a decomposition reaction and breaks down (decomposes) into H_2O and CO_2 (carbon dioxide), which then evolves from the solution.

(Choice A) The H^+ ions in solution have lower electronegativity than the oxygen atoms and cannot retain the electrons necessary to form the $\text{H}-\text{H}$ bond present in hydrogen gas (H_2).

(Choice C) As seen in the Lewis structure of carbonic acid (H_2CO_3), the oxygen atoms are separately bonded to the carbon atom and cannot readily associate to form oxygen gas (O_2). Instead, an acidic proton (H^+) is easily transferred in solution between the two hydroxyl groups to form a good leaving group (a water molecule), which facilitates the breakdown of the molecule.

(Choice D) MgO is a water-soluble, ionic solid that does not form in the described double replacement reaction. The CO_3^{2-} anion does not fragment into O^{2-} anions that could bind with Mg^{+2} ions.

Educational objective:

Double replacement reactions involve the exchange of the bonding partners of two reactant compounds, forming two new product compounds.

Decomposition reactions involve a single compound breaking down into two or more new compounds. Understanding reaction types and bonding patterns allows the prediction of reaction products.

Time spent:0 sec

Subject:
General Chemistry

QID:401760

System:
Interactions of Chemical Substances